



Article

Fake thumbs in play: A large-scale exploration of false amplification and false diminution in online news comment spaces

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Abstract

This study explores how disinformation can dampen general users' expressions of opinion online. In the context of a proven disinformation case in South Korea, this study analyzes externally validated click-logs of 1389 fake accounts and more than a million logs of 45,769 general users in a highly popular web portal. Findings show that the inflated visibility of anti-governmental opinions in the manipulated comment space was incongruent with the overall political tone that general users had spontaneously encountered from the broader media ecosystem beyond the manipulated space. Subsequently, this opinion "climate" incongruence decreased the likelihood of commenting in the manipulated space. The study concludes that false amplification (of

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the opinions that the manipulators promote) and false diminution (of general users' political expressions) work in tandem to create a distorted opinion environment.

Keywords

Click fraud, computational propaganda, disinformation, fake engagement, false amplification, influence operation, opinion manipulation, political astroturfing, political bots, Spiral of Silence

Many of the practices found in digital space are transparent, but covert activities do occur that lie on the boundary of legitimacy. Of recent concern are computational propagandas that employ bot-assisted disinformation activities to manipulate the salience of political opinions in networked online spaces (Woolley and Howard, 2018). While efforts to influence public opinion have always been part of political strategies, the social ramifications of disinformation in today's digital realm have become so far-reaching that they raise serious challenges to democracy. As of 2020, computational propaganda operations have been witnessed in 81 countries, a jump from 28 in 2017, in the forms of both foreign operations and domestic campaigns (Bradshaw et al., 2021).

The current study expands disinformation and computational propaganda research by examining a *dampening* effect of bot-assisted disinformation operations on the online public's discursive participation. Specifically, this study focuses on the creation of fake engagements (e.g. liking, sharing, up/down-voting) as a disinformation technique. Existing research has paid attention mostly to fake *content* creation (e.g. fake news), yet fake *engagements* creation is another important means to amplify the visibility of certain messages. Click frauds have long been an unwelcome element of digital labor markets (Wilbur and Zhu, 2009), and bot-assisted creation of fake engagements has manifested the interplay between human masterminds and automated agents, becoming integral to computational propaganda (Bradshaw and Howard, 2017). Despite its importance, few studies have examined fake engagements due to the dearth of validated data sources. A dataset that discerns fake accounts' click speech activities (Pang et al., 2016) has seldom been accessible to academic communities. Instead, the majority of disinformation studies have relied on content and textual cues, based on which they reverse-engineered manipulative activities, mostly in Twitter (Rauchfleisch and Kaiser, 2020). The current study leverages a unique non-Twitter dataset (i.e. a web portal) that warrants measurement validity of fake engagement activities. The granularity of our data, that includes manipulators' bot-assisted clicking activities, offers a valuable opportunity to investigate the effect of fake engagements on public users' online behaviors.

To offer a theory-driven analytical framework, this study expands the Spiral of Silence (SoS) theory (Noelle-Neumann, 1993). The original SoS theory maintains that the incongruence between one's own opinion and the perceived winning opinion in a given discussion environment results in the reluctance to speak out. We modify the theory's premise by proposing an alternative conceptualization of incongruence, that is, the incongruence stemming from which opinion is winning, that in the manipulated discussion environment

or that in the broader media environment that users are spontaneously exposed to. We then estimate the likelihood of an individual speaking out (or not speaking out) as a function of the incongruence between the two opinion climates.

Unlike a motivational approach taken by the majority of SoS studies (and media and communication research overall), this study takes a behavioral approach by observing real-time log activities of both general users and externally validated manipulator's bot accounts. The empirical context of this study is the "Druking" case of 2018, one of the most notorious opinion rigging scandals in South Korea. This scandal epitomizes an automated creation of fake engagements (e.g. clicks, likes), which has been observed across various digital and social platforms in different disinformation contexts (Bradshaw et al., 2021). By examining the Druking case as an example of a fake engagement operation, this study broadens empirical disinformation research beyond a content-centric approach and outside of Western democracy contexts.

Literature review

Computational propaganda and disinformation research

Digital platforms have become dominant outlets for political conversations in network societies. It has served as an important democratic vehicle that has diversified political talk (Brundidge, 2010) and promoted civic participation and social change (Castells, 2015). That said, recent networked political conversations have sometimes gone wrong with polarization (e.g. Bakshy et al., 2015), incivility (e.g. Kwon and Gruzd, 2017; Coe et al., 2014; Song et al., 2021), and especially the rise of computational propaganda.

Computational propaganda is a form of disinformation that uses "algorithms, automation, and human curation" to "manufacture consensus or to otherwise give the illusion of general support for a (perhaps controversial) political idea or policy, with the goal of creating a bandwagon effect" (Woolley and Howard, 2018: 4). These practices have been ideologically or politically imbued (Ratkiewicz et al., 2011) and have occurred not as individual trolling but as an organized team effort (Bradshaw et al., 2021). Crucially, computational propaganda leverages "automation, scalability, and anonymity" of technological means to rapidly inflate an intended political message (Woolley and Howard, 2018: 5).

Substantial academic effort has been put into the detection of fake accounts (e.g. Davis et al., 2016; Varol et al., 2017; Williamson and Scrofani, 2019) and false information diffusion (e.g. Bastos and Mercea, 2019; Grinberg et al., 2019; Shin et al., 2017; Vosoughi et al., 2018) by analyzing large bodies of digital data. Yet, several gaps exist in the current literature. First, a predominant portion of systematic disinformation research has centered on a single platform, Twitter, and a handful of political events of Western societies. As computational propaganda has become a global trend, there has been a call for broadening investigation beyond Twitter and Western democracies (e.g. Bradshaw and Howard, 2017). However, difficulties in obtaining high-resolution data have hindered the diversifying of cases.

Second, existing data-driven research has enriched a technical understanding of disinformation flow, yet it has paid relatively little attention to impact on general

audience's subsequent informational behaviors. Particularly, the effects of computational propaganda on the message recipients' discursive participation, a cognitive cornerstone of democracy (Cook and Carpini, 2009), have been underexplored. To our knowledge, a few Russian disinformation studies have taken a recipient-centric approach to examine the disinformation effect on users' political attitudes (Bail et al., 2020) and political and network characteristics of retweeters of the disinformation messages (Badawy et al., 2019). However, none have looked at user-behavioral outcomes.

Finally, most disinformation research has centered on detecting and analyzing fake content creation. Reverse-engineering has been an especially useful approach to predict nonorganic activities from content features, profiles, and network characteristics. Some studies have relied on deleted Twitter accounts to trace suspicious tweets (e.g. Bastos and Mercea, 2019). Such proxy representation of disinformation activities is understandable because obtaining ground-truth data is difficult in this research domain. Despite practical utility, however, the "dirty" nature of big data has often failed to disentangle manipulator activities from the activities of laymen users who unwittingly engaged in propagating false messages (Kwon et al., 2018; Salganik, 2019: 39). More importantly, reverse-engineering primarily relies on textual cues, for which disinformation research has inevitably become content-centric. While the production of fake content is a notable problem, equally problematic is the creation of false *engagements* to inflate the visibility of existing content. Disinformation actors across various platforms and different geopolitical contexts have commonly employed human labor and automation to generate fake engagements (Bradshaw et al., 2021), but this has not yet been fully studied because of the rarity of data.

False amplification, false diminution, and SoS

Successful computational propagandas give the impression to the audience that the promoted political stance is the majority one, which in turn engages like-minded audiences into the dissemination process. While this process, called "false amplification" (Freelon et al., 2020: 4), is initiated by manipulators, it eventually becomes far-flung by a bandwagon effect on real people that maintain an ideological stance similar to that of the manipulator (Bail et al., 2020; Bastos and Mercea, 2019).

A flip side of false amplification is *false diminution*. False amplification of a particular position simultaneously sends a (wrong) signal that alternative views are losing ground when, in fact, they may not be. While studies have attested that manipulative activities have mobilized like-minded audiences into the amplification process (Bail et al., 2020; Freelon et al., 2020), it is an open question as to how the manipulation affects general users who are *not* categorized as the like-minded. We conjecture that false diminution may occur with them.

This conjecture is drawn from the SoS theory (Noelle-Neumann, 1993). It contends that humans innately fear being ostracized, and this fear can prevent people from speaking out their opinions about controversial issues. While some studies have conceptualized fear of isolation as a personal attribute inherent in the "psyche of the individual" (Hayes et al., 2013: 442), most SoS research has focused on an exogenous opinion

environment that invokes fear of isolation. When an individual's beliefs or ideas are at odds with the "climate" of opinions in a given situation, they become hesitant to speak out due to the fear of disapproval (Hayes et al., 2013; Ho and McLeod, 2008; Moy et al., 2001; Neuwirth et al., 2007; Noelle-Neumann, 1993).

A recent meta-analysis of SoS research suggested that the relationship between the opinion climate perception and the willingness to express an opinion has held significant in online discussion settings (Matthes et al., 2018), in both social networking sites (e.g. Kwon et al., 2015; Hampton et al., 2017; Matthes et al., 2018), and more anonymous conditions (Nekmat and Gonzenbach, 2013; Zerback and Fawzi, 2017). That being said, most of these studies have examined a self-reported perception of opinion climate in a single digital space (Matthes et al., 2018), while in reality, online users encounter multiple opinion climates simultaneously, or otherwise selectively customize their opinion environs. Schulz and Roessler (2012) called it a "subjective" (p. 346) creation of opinion climate, arguing that diversified digital news sources have allowed users to create a repertoire of ideological notions, which could take an entirely consonant, pluralistic, or even dissonant tone. Given that these choices are made *at will*, questions have arisen as to whether fear of isolation is the only psychological mechanism underlying silence in complex opinion environments (Sohn, 2022), and how the SoS effect would unfold if the exposure to multiple opinion environments are taken into account (Schulz and Roessler, 2012; Sohn, 2022).

Recently, Sohn (2022) has demonstrated that the existence of multiple opinion climates could complicate the process of SoS. Using agent-based simulation modeling, Sohn (2022) showed the interaction patterns between local opinion climates formed via interpersonal networks and a range of diversity in media opinion climates, finding that local opinion climates buffer the effect of media opinion climate unless media opinion was completely uniform.

Such interaction effect between multiple opinion climates is particularly relevant to studying disinformation effects on discursive participation. On one hand, users encounter a distorted opinion climate that the manipulator contaminated. This exposure to the distorted opinion environment can induce "false consensus effect" in that users wrongly believe that manipulator-promoting position is what the majority agrees with (Sohn, 2022: 143). On the other hand, users subjectively construct a broader perception of opinion climate outside of the polluted space via the consumption of plural news sources. Shaping a broader opinion climate depends on the user's overall news diet, and the manipulation that occurs in a single channel (or even a few) may or may not have much influence.

This duality renders a unique conceptualization of incongruence. In the original SoS theoretical framework, opinion incongruence has referred to the discrepancy between one's own thought and the perceived majority opinion in a given situation (Noelle-Neumann, 1993). In the disinformation context, another type of incongruence can be defined as a gap between the manipulated space and a naturally shaped informational scene. Importantly, this opinion "climate" incongruence can occur to not only minority opinion holders (i.e. who disagrees with the opinions inflated by the manipulator) but also moderate or undecided opinion holders, who may not agree with the manipulator's message but not necessarily oppose it, either. Furthermore, the operationalization of

opinion climate incongruence relies not on the self-report of one's own opinion but on the types of information that an individual has been exposed to. Therefore, it offers a methodologically viable option for behavioral analysis.

If the opinion climate incongruence engenders a silencing effect on ordinary users' opinion expression, the process could occur in two steps. First, manipulators would amplify a one-sided political view in a given platform. This amplification should be a false representation of reality, such that it increases the incongruence between the opinion climate within the manipulated space and the outside opinion climate that users would encounter spontaneously.

Hypothesis 1 (H1). Manipulation intensity increases the opinion climate incongruence between the manipulated space and the broader media environment outside the manipulated space.

Next, according to the original SoS theory, opinion incongruence should in turn engender fear of isolation, which causes minority opinion holders to become silent. In a disinformation context, however, manipulation creates the incongruence *between opinion climates*, and this incongruence can occur not only in minority opinion holders but anyone. Therefore, an underlying psychological mechanism may not necessarily be fear of isolation. For example, people can become reluctant to speak out due to "Gresham's law" of political information, which refers to "the less credible source of information driving more credible sources out of consideration" (Boudreau, 2013: 208). Even when the source of information is inaccurate or not entirely credible, an exposure to conflicting information can induce "citizens to either make [political] decisions against their interests or *forgo participation entirely*" (Boudreau, 2013: 207, *italics added*). The incongruent opinion climates are analogous to Gresham's law of political information. That is, the exposure to incongruent opinion climates may signal conflicting information as to which opinion is winning, which in turn makes users forgo discursive participation even when they see that the manipulated opinion space contains less credible messages.

Meanwhile, online users are exposed to the manipulated opinion condition, not in isolation but within a broader informational ecosystem. Accordingly, the exposure to the manipulated opinion climate (MOC) would not discourage one's willingness of discursive participation because the subjectively created opinion climate that exists outside the manipulated space could buffer the impact of false representation. Figure 1 juxtaposes the concept of opinion incongruence in the original SoS theory and opinion climate incongruence that this study proposes in a disinformation context. Considering that the silencing effect of opinion climate incongruence has not been tested previously, we propose this research question:

Research Question 1 (RQ1). How does opinion climate incongruence between the manipulated space and the broader media environment affect a user's likelihood of commenting?

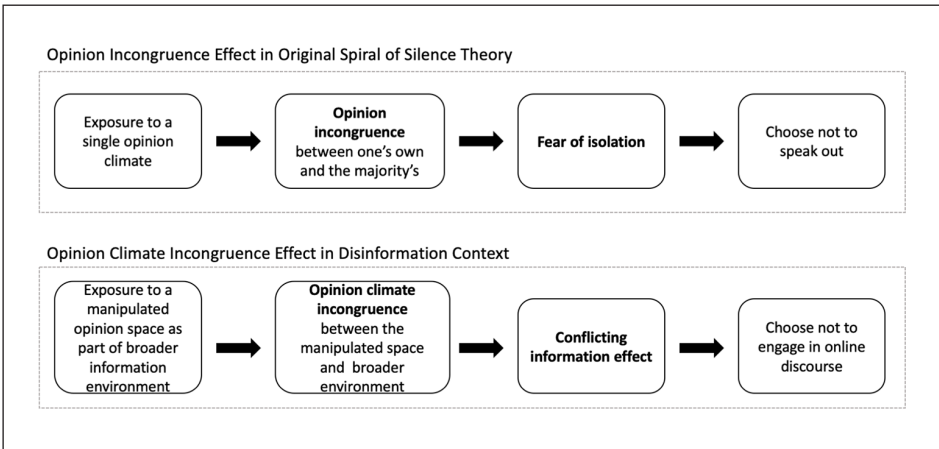


Figure 1. The original Spiral of Silence process versus the modified process in disinformation context.

Fake engagement as a disinformation operation: the 2018 Druking scandal

The current study examines an infamous South Korean disinformation operation in 2018 known as “Druking.” Druking was a screen name of the leader of the operation team who was a popular blogger while covertly having founded a shadow company that ran illegitimate political campaigns online. The company began its operations during the 2017 South Korean presidential election campaign. While their initial stance was pro-Democratic Party (the ruling party at the time), Druking shifted their position to anti-government in 2018, allegedly “in retaliation for the [Party’s] refusal to grant special favors [for jobs at the presidential office]” (Korean Broadcasting System, 2018). The Druking team was arrested in 2018 for opinion rigging.

Druking mainly operated news comment spaces in major Korean portal sites. Internet portal sites are dominant news venues for South Koreans (Newman et al., 2019). News comment spaces in portal sites have been an integral part of South Korean news habits. A nationwide survey reported that 70% of respondents read other users’ news comments in portal sites during the past week; 21% posted comments for themselves; and 31% enacted click speech (e.g. liking) in response to others’ comments (Korea Press Foundation, 2018).

The Druking case is an example of an automated fake engagement operation. Specifically, Druking used an automation tool called “KingCrab,” which generated massive upvotes or downvotes to influence the ranking of comments. The comment ranking mattered because the top-ranked comments could gain disproportionately higher visibility than other comments. Automated creation of fake engagements is a globally used computational propaganda tactic, witnessed in 57 countries as of 2020, according to the Oxford Internet Institute (Bradshaw et al., 2021).

Research design

A case site: a news comment section

The site of interest was a news comment section of a political news story which was hosted in a leading portal site, one of the most popular gateways for South Koreans to access news (Newman et al., 2019). The focal news story, first released on 17 January 2018, at 22:00 (KST), was about the South and North Korean government having agreed to pursue a unified women's ice hockey team for the 2018 Winter Olympics. The South–North relationship has been a persistent partisan agenda item in South Korea. The government's proposal for the unified team alluded to the government's dovish policy toward the North, and supporters for a hawkish policy were critical of this agreement. Public opinion on this issue was closely along partisan lines: The anti-government camp tended to oppose the team, while the pro-government camp was supportive (Sun, 2018). We chose this news story because this topic was later revealed as one that Druking operated on most actively.

The focal news story rapidly garnered a high volume of comments: 39,827 comments within 26 hours after the news release. Interestingly, all of the comments were authentically created, that is, none were manipulated by the Druking accounts. Instead, Druking primarily aimed at creating fake votes. Within the 26-hour window, a total of 1389 Druking's bot accounts acted 56,059 times to cast upvotes or downvotes to shuffle the orders of targeted comments. We identified the Druking accounts based on the verified records that law enforcement provided. We also obtained from the portal company each general user's demographic information and the 3-day news clicks history prior to the release of the focal news story. The platform company de-identified data pertinent to general users before they allowed us access. For ethical considerations, we consulted with the University's Institutional Review Board (IRB) and received approval to use the data.

Data slicing

We focused on the first 2 hours following the release of the news story because the top-ranked comments fluctuated the most dynamically during this time and became stabilized after. For example, there were only two comments newly displayed in the top 20 ranks during the next hour. The first 2 hours covered a significant portion of Druking activities (46.82%), as well as general users' authentic activities (52.92%) (Figures 2 and 3).

Users make sense of an opinion trend in a new comment space by reading others' comments. That said, users probably do not read the entire list of comments. Instead, most users may begin reading them from the top and stop reading at a certain point.

Accordingly, we considered only the top-ranked comments when measuring the winning political position in the comment section at time t . Specifically, we sliced the data in three ways to reflect the scenarios of users reading up to the first 10, 15, or 20 comments. At the time this news was released, the platform's default setting was to display the comments by the net-votes (순공감), calculated as the difference between upvotes

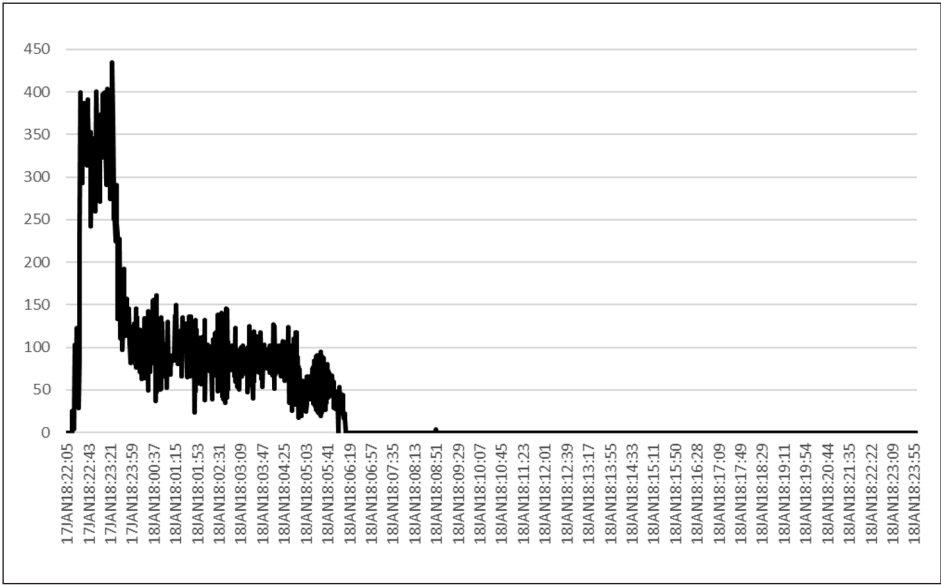


Figure 2. Trend of Druking activities after the news release on 17 January, at 22:00 (KST).

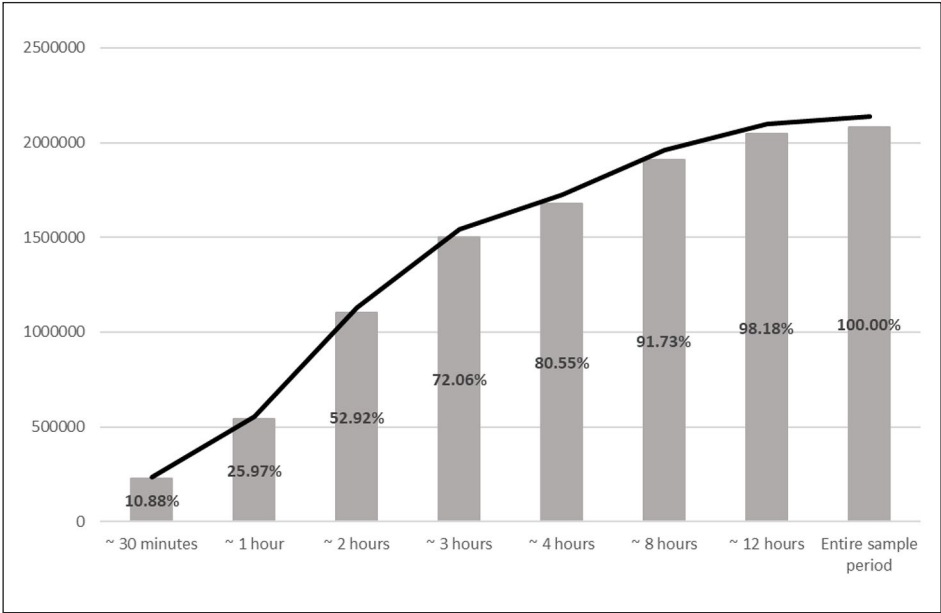


Figure 3. The cumulative number of user activities (the unit of analysis is a user activity per minute).

and downvotes.¹ The comment that earned the highest net-votes was displayed at the top, followed by the comment with the next highest net-votes, and so on.

Measurement

Manipulation intensity. The independent variable for H1 was manipulation intensity, which was measured as the number of Druking's votes divided by the number of total votes that were made across all comments at time t . By construction, the manipulation intensity score lies between 0 and 1.

Opinion climate incongruence. To address H1 and RQ1, we measured the incongruence between the MOC at time t and a user's subjective opinion climate (SOC) constructed outside the manipulated space. Formally, user i 's opinion climate incongruence is defined as

$$\text{Incongruence}_i = \frac{\sum_{k=1}^n I[P_k \neq P_i]}{n} \quad (1)$$

where P_k is the political orientation of a comment k in MOC, P_i is the political orientation of a user i 's SOC, and n is the number of top comments shown in MOC. $I(A)$ is the indicator function such that it equals 1 if A is true and 0 otherwise.

We took three steps to derive this variable. First, to capture MOC, we requested 12 graduate students to code the political orientation of each of the top-ranked comments, based on the 5-point scale, 1 = *pro-government*, 2 = *between pro-government and neutral*, 3 = *neutral/pluralistic*, 4 = *between neutral and anti-government*, and 5 = *anti-government*; intraclass correlation coefficient (ICC) = .983.² We then derived P_k by averaging the scores and converting them into a three-category scale, including "pro-government" (i.e. the mean of 12 scores was two or less), "neutral/pluralistic" (i.e. the mean was between two and four), and "anti-government" (i.e. the mean was larger than four).

Next, to quantify P_i , the political orientation of a user i 's SOC, we used the 3-day history of political news pages that each user had clicked before the release of the focal news story. Under the guidance of a journalist and a media scholar, we categorized the publishers of clicked pages into anti-government (conservative), pro-government (liberal), and nonpartisan/pluralistic media outlets. Based on the publishers' political tendency, we defined P_i as "anti-government" if conservative news outlets appeared more frequently in the logs, "pro-government" if liberal news outlets appeared more frequently, and as "neutral/pluralistic" otherwise.

Finally, we computed a user i 's opinion climate incongruence as the proportion of comments (among the top-ranked at time t) that were incongruent with the user's SOC (equation (1)). The range of opinion incongruence scores is between 0 (complete congruence) to 1 (complete incongruence).

Dependent variables. The dependent variable for H1 was the overall incongruence between the MOC and SOC. We call this variable "overall" opinion climate incongruence, given that it was measured per minute by averaging the individual incongruence

scores explained above. The dependent variable to address RQ1 was whether a user i commented on the focal news page at time t ($=1$) or not ($=0$).

Control variables. For time-invariant variables, we added each user's age, gender, and commenting propensity ($=1$ if a user had commented to a political news story in the past 3 days before the release of the focal news story and $=0$ otherwise). We also controlled the political orientation of the SOC ($=1$ if user's SOC was pro-government and $=0$ otherwise). Also controlled were time-varying variables, including (a) the time of the day at which a user visited the focal news page, (b) the tenure (i.e. the interval between a user's visiting time and the time of news release), and (c) the access device ($=1$ if a user visited the focal news page via a mobile device and $=0$ otherwise).

Analytic strategy

We removed the users whose SOC was explicitly anti-government from the analysis, assuming that their viewpoints were presumably consonant with the Druking's position, thus not applicable to the notion of incongruence. After the removal of the consonant users ($n=5995$) and users with missing information ($n=19,921$), we retained a total of 45,769 users for further analysis.

We conducted linear regression models to test H1 and logistic regression models to address RQ1. The unit of analysis for H1 was time t (minutes; $n=115$ for the first 2 hours), and the unit of analysis for RQ1 was user i and time t ($n=1,126,026$ for the first 2 hours).

For robustness checks, we used two modified samples. First, we expanded the time-window to 3 hours. By including one extra hour, we embraced affected users more comprehensively while the variance of top comment compositions was still reasonable for reliable analysis. The 3-hour analysis included 175 time-units (t) and covered 57.43% of Druking and 72.06% of general user activities ($N=1,532,800$).

Second, we focused on mobile users. The majority of users in our dataset (87.6%) accessed the news page via a mobile device. A mobile device displayed the top five comments by default, and the next batch of five comments would appear each time a user clicked the "more comments" button. Meanwhile, a computer screen displayed the top 10 comments by default, and the next batch of 10 each time a user clicked the button. Given that the number of comments displayed per click was different between the two device types, we examined if the need for an extra click entailed different results for the mobile users.

Results

Descriptive results

We report descriptive results based on the observation of the first 2 hours in Table 1. The majority of users who visited the focal news page (78.39%) were between 20 and 49 years of age and male (76.31%). The vast majority of users (97.44%) maintained a neutral/pluralistic SOC, indicating that their selection of political news outlets did not show an

Table 1. Descriptive statistics of variables based on the first 2-hour sample.

User level (<i>n</i> = 45,769)		Mean	SD
Commenting on the focal news (0/1)		0.110	0.314
Commenting in the past 3 days (0/1)		0.282	0.450
		Frequency	%
Age	Missing	17	0.037
	~19	2394	5.231
	20–29	13,627	29.773
	30–39	12,715	27.781
	40–49	9536	20.835
	50–59	5690	12.432
	60–69	1543	3.371
	70~	247	0.540
Gender	Missing	17	0.037
	Female	10,825	23.651
	Male	34,927	76.311
Subjective opinion climate (SOC)	Neutral	44,599	97.444
	Pro-government	1170	2.556
User-time level (<i>N</i> = 1,126,026)		Mean	SD
Opinion Incongruence (0–1)	Top 10	0.860	0.025
	Top 15	0.870	0.035
	Top 20	0.885	0.026
Time of the day		22:50	0.500
Mobile use (0/1)		0.876	0.330
Tenure (lagged click time from the news release, in seconds)		3723.920	1882.420
Time level (<i>n</i> = 115)		Mean	SD
Overall opinion incongruence (0–1)	Top 10	0.857	0.041
	Top 15	0.863	0.048
	Top 20	0.878	0.046
Manipulation intensity (0–1)		0.027	0.015

SD: standard deviation.

obvious partisan affinity. In terms of commenting behaviors, about one-third of the users (28.2%) had commented on any political news during the last 3 days before the release of the focal news story. On the focal news page, a much smaller percentage of users (11%) left a comment at least once during the observed hours,³ and the chance of a user making a comment at *t* was even lower (0.9%).

A total of 42 unique posts were ranked at least momentarily in the top 20 comments. Overall, the opinion climate in the top-ranked comments (MOC) was highly incongruent from SOC, with an incongruence score of 0.857 in the top 10, 0.863 in the top 15, and 0.878 in the top 20. The high incongruence scores were due to the obvious

anti-governmental stance revealed in the top comments, which was in line with Druking's intention. For example, a top-ranked comment that Druking actively promoted by generating upvotes stated,

Whose country is this? Bring to justice by impeaching Moon Jae-in (the president of South Korea at the time), who offers the whole nation as a tribute to North Korea (이게 나라냐? 온 나라를 북한에 떠다 바치는 문재인 정권 탄핵으로 심판하자).

Meanwhile, comments that Druking demoted by generating downvotes tended to be neutral or supportive of the government's decision, for example,

The South-North's unified team is the first-ever achievement in the Olympic history . . . showing the true spirit of Olympic Games for the world peace (남북 단일팀이 성사되면 올림픽 사상최초 . . . 세계 평화 유지라는 올림픽 정신에도 부합되는 것 . . .).

Interestingly, some comments that disappeared from the top ranks included suspicions of disinformation operations, on which Druking aggressively generated downvotes, for example, "Let's urge for an investigation on whether A (the portal site's name) is doing a manipulation operation (A 포탈 조작질 당장 조사하라)."

Hypothesis and research question

H1 posits that manipulation intensity increases opinion climate incongruence. The results showed a significant positive association under the "top 10 comments" scenario that users would read the first 10 comments to determine their perception of MOC ($\beta = 1.068$, $p < .001$). However, the results were *not* significant in the "top 15 comments" and "top 20 comments" scenarios (Table 2).

Regarding RQ1, about the relationship between the opinion climate incongruence and the likelihood of commenting, the results were not significant under the "top 10 comments" scenario. However, the results were significant in the other scenarios, showing that the likelihood of commenting significantly decreased as the incongruence increased in the "top 15 comments" ($\beta = -2.615$, odds ratio = 0.073, $p < .001$) and "top 20 comments" scenario ($\beta = -2.658$, odds ratio = 0.070, $p < .001$) (Table 3).

Among control variables, users who had a pro-governmental SOC were more likely to leave a comment across all scenarios. Likewise, users who had commented on a political news page in the previous 3 days and those who accessed the focal news page via a mobile device were more likely to comment on the focal news page. Male users were more likely to comment than female users. The results of control variables suggest that users' political affinity, commenting propensity, device types, and gender may also be important determinants of commenting behaviors, along with the opinion climate incongruence.

Robustness checks

We expanded the analyses with two modified sampling scopes: by adding one extra hour of observations and by sub-sampling mobile users. To summarize, we found consistent

Table 2. Regression models to estimate the effect of voting manipulation on overall opinion incongruence.

	Top 10		Top 15		Top 20	
	B	SE B	B	SE B	B	SE B
(1) First 2 hours (n = 115)						
Manipulation intensity	1.068***	0.232	−0.022	0.298	0.100	0.280
Constant	0.828***	0.007	0.864***	0.009	0.875***	0.009
	F(1, 113) = 21.231***		F(1, 113) = 0.005		F(1, 113) = 0.127	
	R ² = .158		R ² = .00005		R ² = .001	
(2) First 3 hours (n = 175)						
Manipulation intensity	0.733***	0.188	0.087	0.223	0.034	0.210
Constant	0.846***	0.005	0.856***	0.006	0.880***	0.006
	F(1, 173) = 15.240***		F(1, 173) = 0.153		F(1, 173) = 0.026	
	R ² = .081		R ² = .001		R ² = .0002	
(3) Mobile activities (n = 115)						
Manipulation intensity	1.065***	0.232	−0.024	0.297	0.097	0.280
Constant	0.828***	0.007	0.864***	0.009	0.875***	0.009
	F(1, 113) = 21.057***		F(1, 113) = 0.006		F(1, 113) = 0.120	
	R ² = .157		R ² = .0001		R ² = .001	

Unit of analysis = *t* (minute).
****p* < .001.

results with both alternative samples. For H1, the results with both modified sampling scopes showed that the manipulation intensity was positively associated with the opinion climate incongruence under the scenario of the top 10. At the same time, no significant association existed under the scenarios of the top 15 and 20 comments, consistent with the main results based on 2-hour observations.

Likewise, both modified sampling scopes showed that the opinion climate incongruence was negatively associated with the likelihood of commenting under the scenarios of the top 15 and 20. There was no significant association under the scenarios of the top 10, which was, again, consistent with the 2-hour analysis. The results of most of the control variables also remained consistent.

Discussion and conclusions

Networked democracies have witnessed how computational propaganda infiltrates public minds and compromises the democratic potential of mediated communication. A few recent studies (e.g. Bail et al., 2020; Freelon et al., 2020) have unwrapped the mechanism of false amplification that involved the interplay between disinformation actors and like-minded users. However, a more comprehensive picture of disinformation and computational propaganda machinery necessitates looking at the roles *the rest of the users* would play in the process.

Table 3. Logistic regression models to estimate the effect of opinion incongruence on a user's commenting behavior.

	Top 10			Top 15			Top 20		
	B	SE	e ^B	B	SE	e ^B	B	SE	e ^B
(1) Observations for the first 2 hours (n = 1,126,026)									
Opinion incongruence	1.161	1.362	3.192	-2.615***	0.539	0.073	-2.658***	0.467	0.070
Age ~19	-0.440	1.012	0.644	-0.432	1.012	0.649	-0.413	1.012	0.662
Age 20-29	-0.367	1.011	0.693	-0.356	1.011	0.701	-0.337	1.011	0.714
Age 30-39	-0.242	1.011	0.785	-0.231	1.011	0.794	-0.213	1.011	0.808
Age 40-49	-0.102	1.011	0.903	-0.092	1.011	0.912	-0.075	1.011	0.928
Age 50-59	-0.169	1.011	0.845	-0.158	1.011	0.854	-0.144	1.011	0.866
Age 60-69	0.224	1.012	1.251	0.233	1.012	1.262	0.247	1.012	1.280
Age 70~	-0.220	1.017	0.802	-0.212	1.017	0.809	-0.187	1.017	0.829
Gender (female)	-0.868***	0.023	0.420	-0.867***	0.023	0.420	-0.865***	0.023	0.421
Pro-government SOC	0.515**	0.198	1.674	1.041***	0.105	2.831	0.978***	0.090	2.658
Logged in the first hour	-0.105*	0.047	0.900	-0.001	0.046	0.999	-0.129***	0.039	0.879
Commented in the past	2.432***	0.041	11.378	2.432***	0.041	11.378	2.432***	0.041	11.384
User's tenure (seconds)	-0.00002	0.00001	1.000	-0.00002*	0.00001	1.000	-0.00003**	0.00001	1.000
Mobile devices	0.141***	0.028	1.152	0.141***	0.028	1.151	0.141***	0.028	1.151
Constant	-7.093***	1.542	-3.895***	-3.895***	1.110	-3.737***	-3.737***	1.094	
			$\chi^2(14) = 8785.825^{***}$			$\chi^2(14) = 8807.452^{***}$			
(2) Observations for the first 3 hours (N = 1,532,800)									
Opinion incongruence	2.538	1.312	12.657	-2.731***	0.539	0.065	-2.914***	0.458	0.054
Constant	-8.183***	1.514	-3.700***	-3.700***	1.107	-3.430**	-3.430**	1.087	
			$\chi^2(15) = 11,380.457^{***}$			$\chi^2(15) = 11,400.953^{***}$			
(3) Subsamples with mobile activities (n = 986,474)									
Opinion incongruence	0.760	1.500	2.138	-1.718***	0.603	0.179	-1.965***	0.514	0.140
Constant	-6.606***	1.634	-4.509***	-4.509***	1.134	-4.208***	-4.208***	1.112	
			$\chi^2(13) = 8130.984^{***}$			$\chi^2(13) = 8138.615^{***}$			

SOC: subjective opinion climate. e^B=exponentiated B. Unit of analysis = a user activity at t (minute). Control variable estimates for (2) and (3) are available upon request. *p < .05. **p < .01. ***p < .001.

The current study contends that ordinary users can contribute to a distorted opinion distribution not only by engaging with (thus amplifying) the manipulator voices but also by eschewing from speaking out (thus diminishing) alternative perspectives. Manipulators may achieve their goal to create a “false sense of group consensus” (Ratkiewicz et al., 2011: 249) by having false amplification and diminution work in tandem. At the heart of this study’s premise lies the concept of opinion climate incongruence, which we defined by modifying the original SoS theory.

The empirical case we analyzed was the Druking scandal in South Korea, an example of an automated fake engagements operation. Fake engagements operation is an important disinformation technique but has yet been fully explored due to data rarity. Leveraging a unique data source that contained granular information on bots and general users’ activity logs, this study showed that automated fake engagements operation inflated the visibility of political stance that the manipulator promoted, which in turn resulted in the increased incongruence between the MOC and the SOC that users spontaneously constructed in a broader informational ecosystem.

Interestingly, the evidence of such false amplification was found *only from the top 10 comments* in our study context. In other words, the effect of manipulation was significantly positive when we assumed that the first 10 comments would determine a user’s perception of opinion climate in the manipulated news page. However, no significant finding existed when the top 15 and 20 comments were considered. The nonsignificant results with regard to the top 15 and 20 comments may seem counterintuitive because the overall scores of opinion climate incongruence were kept high across all scenarios. After looking at the results more closely, however, we concluded these findings could hint that not only manipulators but also authentic users partook in the amplification process.

Stated differently, this result suggests a possibility that Druking had operated mainly to set the anti-governmental sentiment in the most visible space (in the top 10 ranks). Yet, the still high level of incongruence in the top 15 and 20 implies that the anti-governmental sentiment spilled over into the next batch of comments, not because of the Druking’s direct interference (given it was statistically nonsignificant), but because of other factors. These other factors could be real users’ engagements that did Druking a favor. That is, a bandwagon effect could have existed among like-minded users. If the logic of a bandwagon effect, however, requires validation of the similarity in the activity patterns between the Druking bots and the like-minded users, this is unfortunately beyond the scope of the current study. Nonetheless, the present findings suggest a direction for future research.

Furthermore, we examined whether the opinion climate incongruence should decrease a user’s commenting. The results showed evidence of false diminution *not in the top 10 comments but in the top 15 and 20 comments*. One way to interpret these results is that users who commented after reading the top 10 comments could be “the hardcore,” highly opinionated individuals that are “in defiance of the threats of isolation” (Noelle-Neumann, 1993: 170). While we cannot verify whether they indeed shared the hardcore’s characteristics, the significant positive effect of pro-government SOC tends to support this point. The 2.56% of users with pro-government SOC took the overt opposite of the manipulator’s position but were more likely to comment irrespective of being the minority opinion holders.

Another possible interpretation pertains to the boundary condition of the SoS effect. For example, a threshold for the number of comments that a user needs to read to feel discomfort could be more than 10. This interpretation is reminiscent of Ordoñez and Nekmat's (2019) study on the tipping point of SoS effect. They (Ordoñez and Nekmat, 2019) pointed out that an experimental setting with the fixed number of displayed comments could obscure generalizability, leaving it unanswered whether the tipping point would change as a function of the number of displayed comments. In this sense, this study complements Ordoñez and Nekmat (2019) by showing that the number of displayed comments affects behavioral outcomes.

One important contribution of this study is the conceptualization of opinion climate incongruence (i.e. between MOC and SOC). This concept departs from the original SoS theory that focused on the perceptual difference between an individual's own opinion and the winning opinion in a discussion environment. Meanwhile, the concept of opinion climate incongruence in this study centers on the use of multiple informational sources, under the assumption that information consumption leads to and arises from one's perception of a winning political position. Although inferring an individual's political perception from information uses has been a common practice in digital data-driven research (e.g. Messing et al., 2020), it nonetheless is a limitation that we could not validate the assumption without having each user's perception measured. In particular, the 3-day history of news views was narrow to infer SOC most reliably. However, it is reasonable and ethical from a data privacy perspective that the platform company allowed researchers access to the user data of a limited time-window.

The original SoS theory focuses primarily on the minority opinion holders, whose position would be clearly opposite of the perceived majority opinion. In contrast, most users (97.44%) in our dataset showed a neutral/pluralistic SOC. This composition may reflect the true reality of the opinion distribution wherein users tend to maintain moderate or mixed political views. Considering that our findings pertained to these moderate users, we suggest that a conflicting information effect, rather than fear of isolation, could better explain why users chose not to leave a comment. The extreme partisanship exhibited in MOC could puzzle users who have been accustomed to the moderate or pluralistic political tones in other news sources. Whether it is due to fear of isolation or conflicting information, the danger of opinion manipulation seems to lie not only in falsely amplifying a particular position but also in diminishing moderate voices into silence.

In conclusion, this study advances computational propaganda and disinformation research by showing how fake engagement operations can dampen moderate public voices in online discursive spaces. The findings of this study may offer managerial insights into how to design online comment spaces. A vote-based ordering of comments is an easy solution for platform producers, but it is likewise one of the easiest targets for manipulation. Prioritizing high-quality and deliberative comments and enhancing their visibility in the ranking system can nurture a healthier and pluralistic discursive culture, although it is a difficult task to define what constitutes a high-quality opinion. Platform producers may delve further into what defines the deliberative quality of comments and how to detect and quantify them into ranking algorithms.

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Notes

1. The comment ordering rule has changed since the Druking scandal. Currently, publishers can choose one of two options, by the net-votes or by recency, for the default display of comments under their news articles.
2. Type A intraclass coefficient using an absolute agreement definition; the reported value is based on the mean. Intraclass correlation coefficient (ICC)=.824 for a single-rater measure.
3. While a rare event could be a potential issue, the large number of "1" in our dataset legitimizes the use of a logistic regression model. See Wang (2020).

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